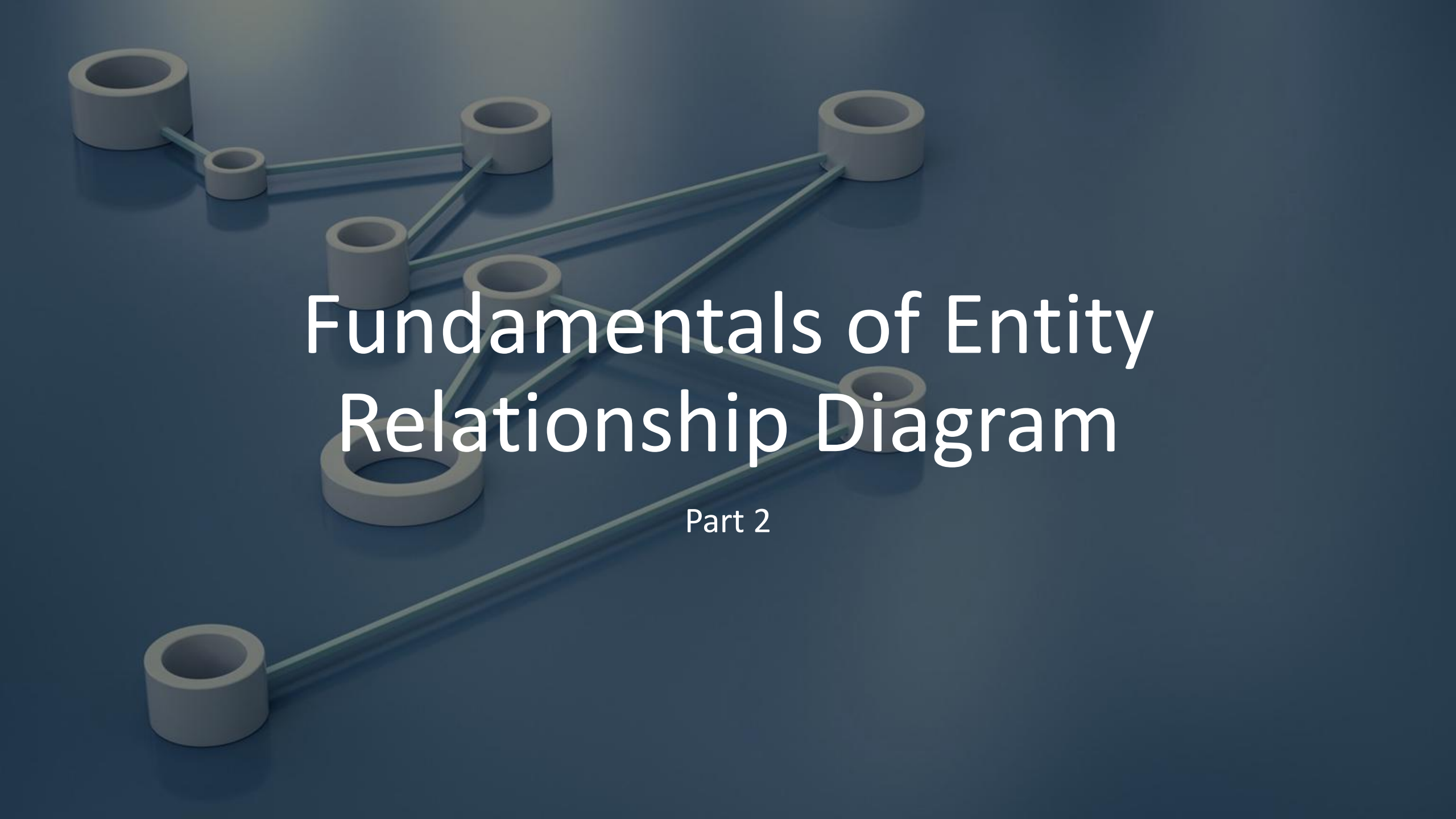


A 3D visualization of an Entity-Relationship Diagram (ERD) on a dark blue background. The diagram consists of several light gray cylindrical nodes of varying sizes connected by thin, light blue lines. The nodes are arranged in a network, with some acting as central hubs and others as peripheral nodes. The lines represent the relationships between these entities. The overall structure is complex, showing multiple interconnected paths and clusters.

Fundamentals of Entity Relationship Diagram

Part 2

Objectives

- Know the 2 common types of notations used in ER Modeling.
- Get to know additional details of strong and weak entities.

Additional Information

Name _____

Signature _____

Date _____

Notations

2 commonly used notations are:

- Chen's Notation
- Crow's Foot Notation

Chen's Notation

- **Chen ER diagram** is an entity-relationship diagram notation that shows entities, attributes, and relationships within a data model.
- It is a useful method for designing conceptual models.
- It depicts the entity attributes and relationships in separate boxes, connected to the entities.

Chen's Notation

- The **Chen notation** was created by Peter Chen, a computer scientist who specialized in database theory.
- He is most notable for his work on the entity-relationship model, which he first published in 1976.

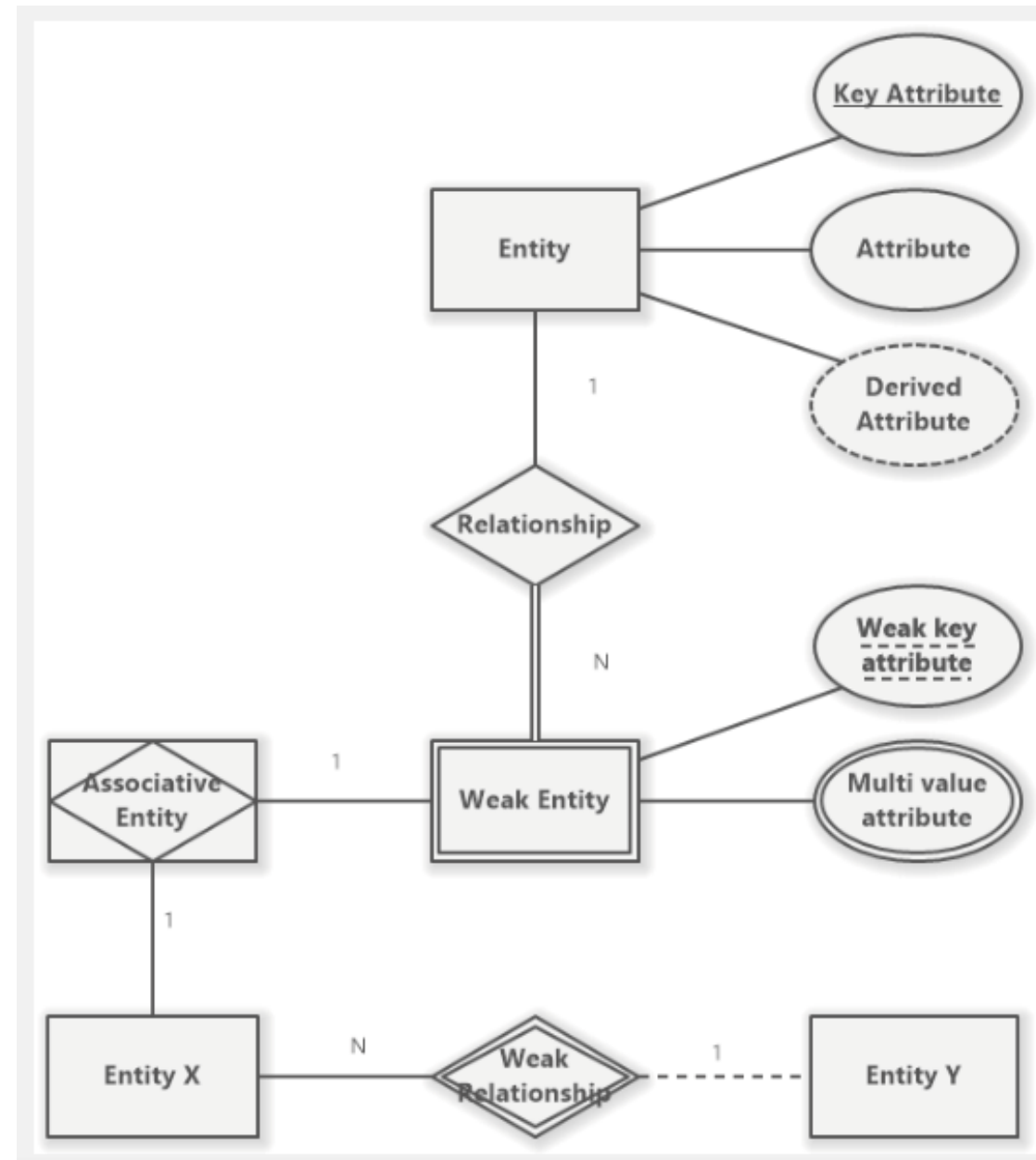
Cardinality

- **One-to-one** (1:1)
- **One-to-many** (1:N)
- **Many-to-one** (N:1)
- **Many-to-many** (M:N)

Chen's Notation using Symbols – an example



Chen's Notation using Symbols: An Example



Crow's Foot Notation

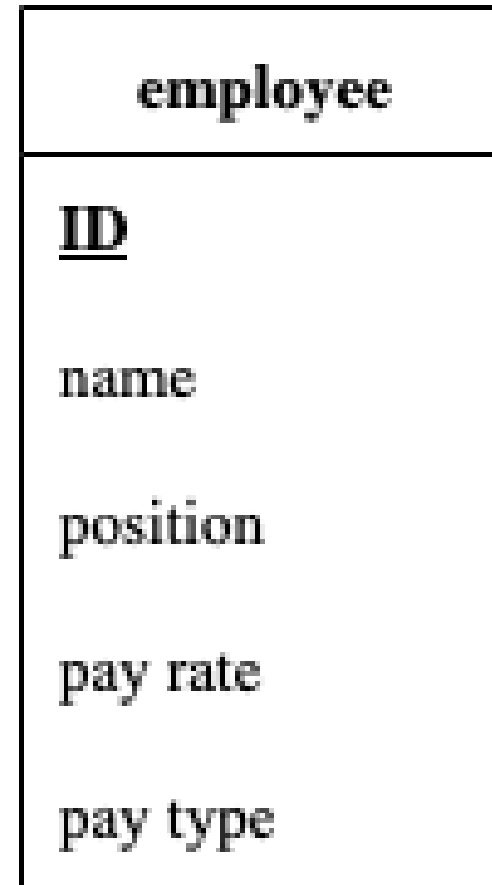
- “Crow's foot” is the nickname given to a family of modeling notations that originated in a paper by Gordon Everest in 1976.
- The notation has been widely adopted, expanded, and modified since.
- The name comes from the symbol used to represent the “many” cardinality in relationships, which resembles a crow's foot
- Everest originally called the symbol an “inverted arrow”, and later a “fork”.

Crow's Foot Notation

- Entities are represented with simple rectangles.
- Entities can be labeled simply with a name for a higher level of abstraction, or they can contain a name and a list of attributes.
- Key attributes can be indicated in various ways;

Crow's Foot Notation

- In crow's foot notation, an entity is represented by a rectangle, with its name on the top.
- The name is singular (entity) rather than plural (entities).
- An attribute is a property that describes a particular entity.

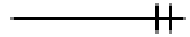


} **attributes**

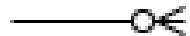
Crow's Foot Notation- Cardinalities



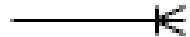
Zero or one



One exactly



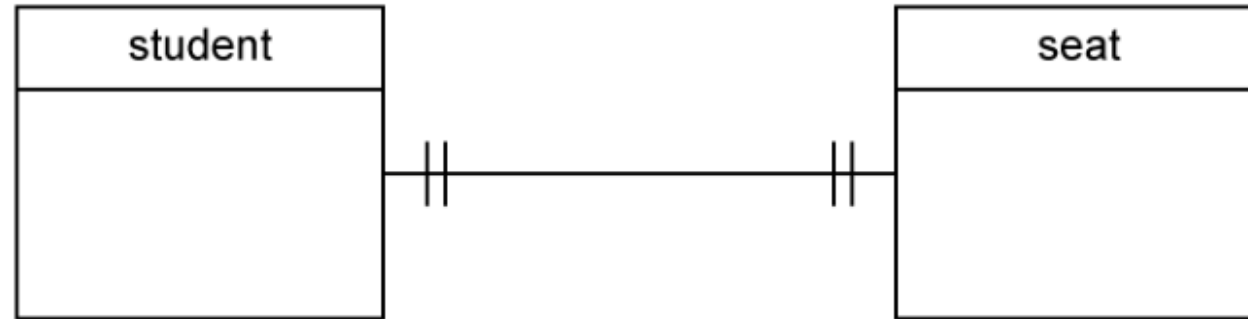
Zero or more



One or more

Cardinalities-Representation

- One-to-one

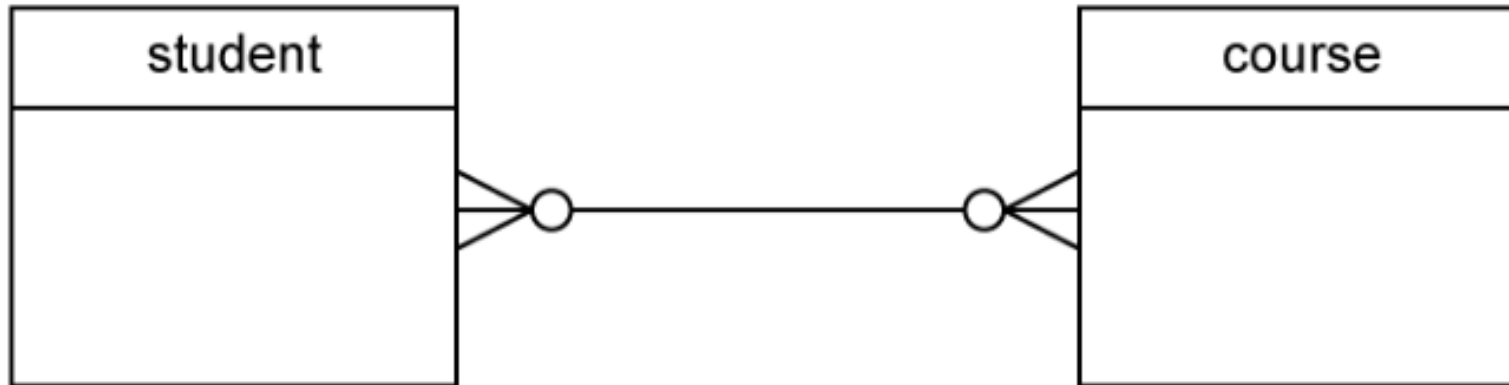


- One-to-many

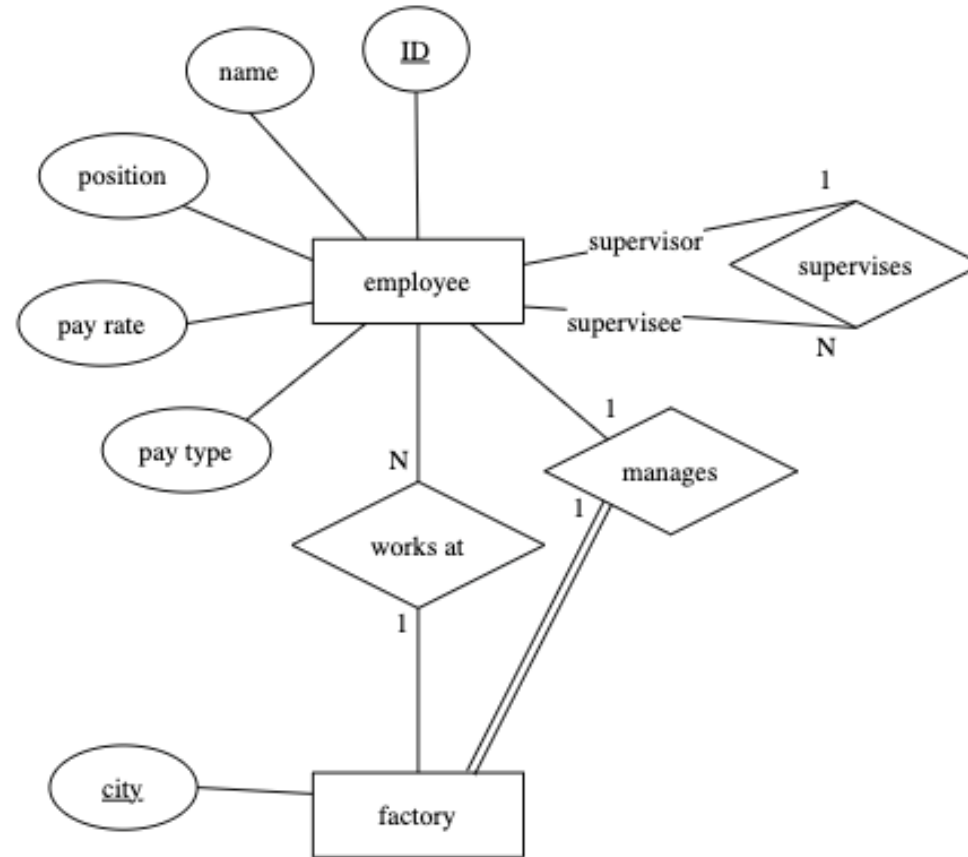
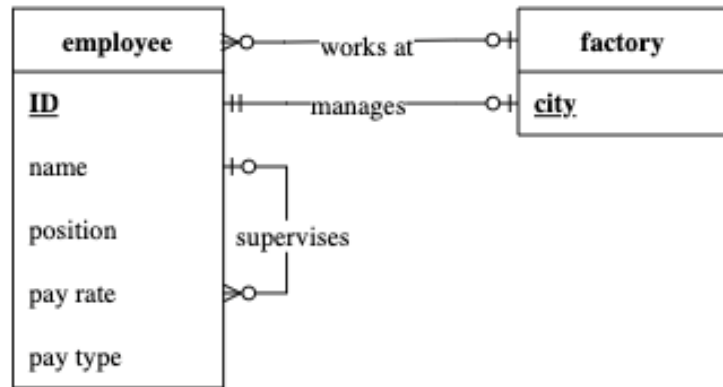


Cardinalities-Representation

- Many-to-many



Crow's Foot vs. Chen Notation



Weak Entities

- A **weak entity** is an entity whose instances (rows) cannot exist in the database without the existence of an instance of another entity (a row in another table).

Strong Entities

- An entity that is not weak is called a strong entity.
- Instances of a strong entity can exist in the database independently.

Types of Weak Entities

- **ID-Dependent Weak Entity-** cannot exist without its parent entity.
- For ID-Dependent weak entities, this requirement is enforced by using a composite primary key in the weak entity.
- The first part of the composite key is for the strong entity
- The second part of the composite key is for the weak entity itself.

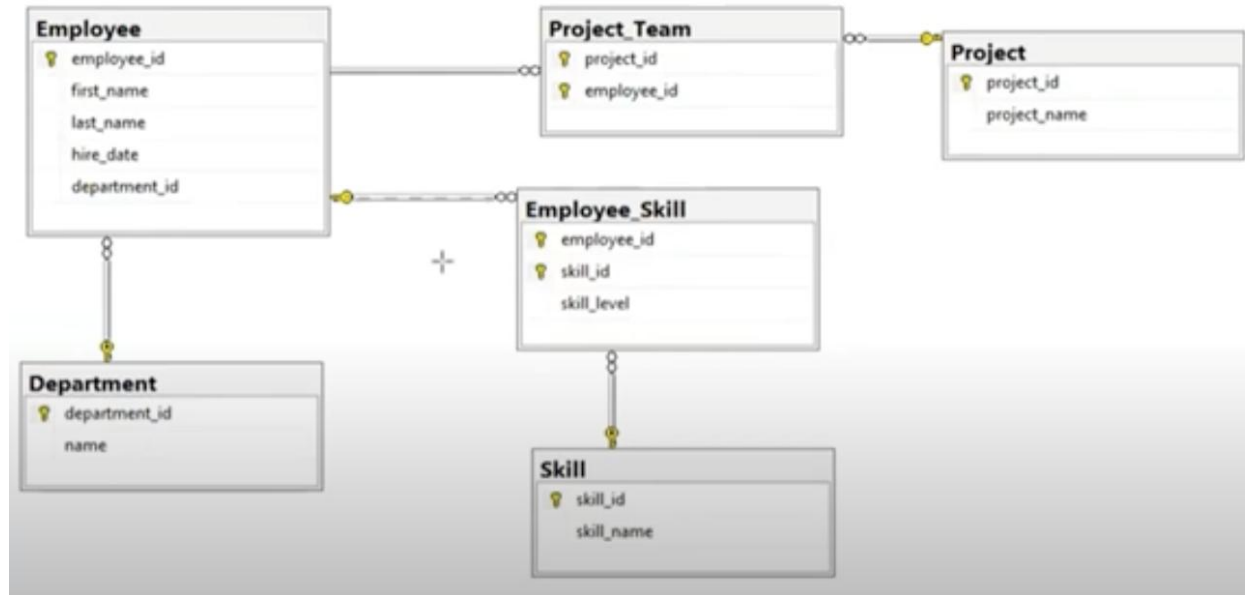
Types of Weak Entities

Non-ID Dependent weak entity has its own ID but has foreign key.

Types of Weak Entities

What are the strong entities?

The strong entities are:
Department, Skill, Project



What are the weak entities?

The weak entities are employee,
Project team, Employee skill

References:

- <https://vertabelo.com/blog/crow-s-foot-notation/>
- <https://miro.com/diagramming/crows-foot-notation-in-er-diagrams/>
- <https://www.softwareideas.net>
- <https://www.gleek.io/blog/crows-foot-chen>